



Using the fx-CG50 for statistics

Task A: Numerical summaries of data

1) This frequency table shows the star rating received by a restaurant over its opening weekend.

Star rating	Frequency
1	12
2	18
3	21
4	38
5	45

a) What was the mean star rating?

3.64... stars (which rounds to 4)

b) What was the median star rating?

4 stars

2) The frequency table shows the number of bedrooms in different houses within an area of a town.

Bedrooms	Frequency
1	20
2	93
3	54
4	33

a) What is the mean number of bedrooms?

2.5 bedrooms (which rounds to 3)

b) What is the median number of bedrooms?

2 bedrooms

3) The time it takes for a sample of pupils to travel to school is collected in this grouped frequency table.

Time taken to travel to school (in minutes)	Frequency (<i>f</i>)
$0 < t \leq 4$	28
$4 < t \leq 10$	23
$10 < t \leq 16$	70
$16 < t \leq 26$	52
$26 < t \leq 40$	30
$40 < t \leq 60$	13

a) What is the estimated mean travel time (to the nearest minute)?

18 minutes

b) What is the estimated median travel time (to the nearest minute)?

13 minutes

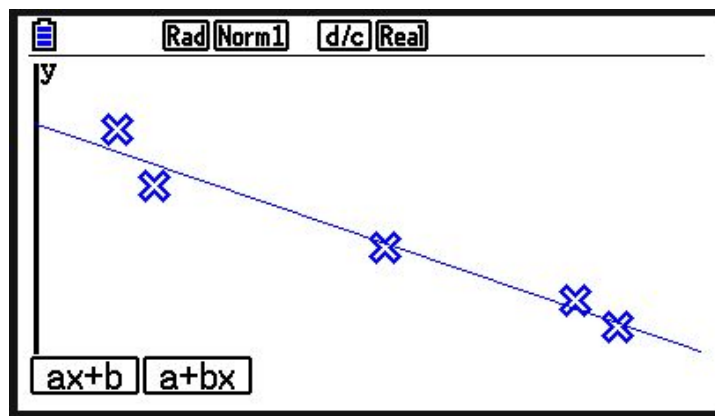


Task B: Graphical representations of data

1) The table shows weather data from Bradford, for five months in 2021.

What does this data suggest about the weather? Justify your answer.

Month	Total rainfall (mm)	Total sunshine (hours)
Feb	100	66
Apr	16	216
Jun	22	173
Aug	61	126
Oct	93	86

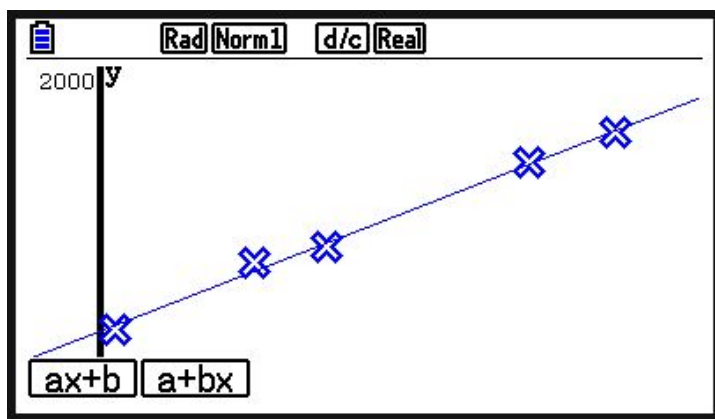


There is negative correlation. More rain means less sunshine.

2) Jacob runs a series of long-distance races. For each race, he records his time and the elevation of the course.

What does this data suggest about elevation and the time taken? Justify your answer.

elevation (m)	time (s)
51	1756
4	1644
35	1735
115	1911
96	1870



There is positive correlation. The greater the elevation, the more time it takes to run the course.

